higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIALS
NQF LEVEL 2

1 MARCH 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1 1.1.1 The ease with which a material catches fire

1.1.2 Density of material relative to density of water

1.1.3 Property of material showing how easily it allows the flow of electricity (flow of electrons)

(3 × 2) (6)

1.2
- More close adherence to specifications.
- More consistent in quality with less wastage.
- Safer to use.
- Stronger because of better matter material
- More healthy working and living environment
- Building lasts longer (Any 3 × 1) (3)

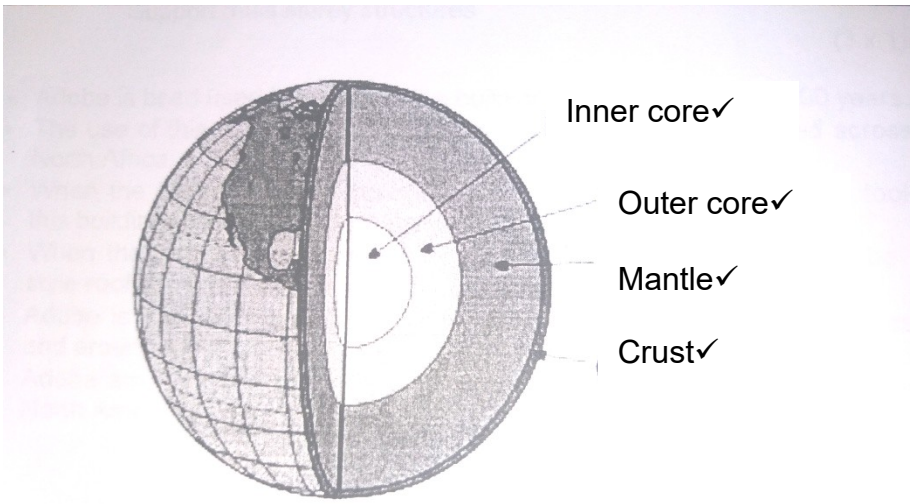
1.3 Recycling is the process whereby materials are reused for another purpose or to produce the same product from which the waste was produced. (1)

[10]

QUESTION 2

2.1 Geological time refers to the timing of events on earth,✓ usually measured in million years.✓ (2)

2.2



(4)

2.3 Metamorphic rocks are formed from igneous or sedimentary rocks✓ by a process called metamorphism.✓ Sedimentary and Igneous rocks are chemically altered under high pressures and temperatures✓ deep inside the earth to form metamorphic rocks.✓ (4)

[10]

QUESTION3

3.1 Soil compaction means pressing soil particles together,✓ reducing the air and water filling the gaps between the particles.✓ (2)

3.2
- Compacting soil increases its density and ability to support heavy loads.
- It decreases the permeability of the soil and the amount of settlement under a heavy load.
- When compacted the soil reduces in volume (in some cases soil has to be ordered to backfill a certain volume of space). (Any 2 × 1) (2)

3.3 Rubber-tyre rollers (1)

3.4
- Easily found
- Cheap
- Bricks easily made
- Durable building material
- Good temperature control as each brick absorbs heat by day and releases it by night
- Sticks together and does not crumble in rainy climates (Any 2 × 1) (2)

3.5
- Active (expansive) soil
- Collapsing soil
- Dolomite
- Polluted soil
- Salination
- Soft clays
- Clay soil (Any 2 × 1) (2)

3.6
- Raft foundation
- Wide strip footings
- Pile foundation (Any ONE) (1)

[10]

QUESTION 4

4.1 Soak the brick in water✓ before use to reduce the amount of moisture✓ it absorbs from the mortar.✓ (3)

4.2
- Use a mortar mix that contains materials that can accommodate expansion.
- Build walls that are not too long.
- If a long wall is required, build control joints in preparation for any expansion that may occur.
- Moisten the bricks before laying them. (Any 3 × 1) (3)

4.3 Vanadium stains are a yellow-green✓ discolouration on the surface of a clay brick.✓ (2)

4.4
- Preclean the large chunks of mortar that would have dried after 24 hours with the wooden chisel.
- Apply hydrochloric acid using a broom. Allow it to soak the mortar and dissolve the unwanted cement.
- Wash off the acid after all stains are removed. (3)

4.5 4.5.1 Bunker ash is a byproduct✓ of the coal-fired heating process.✓

4.5.2 Crushed burnt clay is a waste product✓ that occurs in the brick-making process where certain bricks are rejected.✓

(2 × 2) (4)
[15]

QUESTION 5

5.1
5.1.1 True
5.1.2 True
5.1.3 True
5.1.4 False
5.1.5 True

(5 × 1) (5)

5.2
- Water-cement ratio
- Aging
- Hydration
- Drying out (Any 3 × 1) (3)

5.3 Stones must be strong enough to resist wearing away during compaction✓ so that it can transmit the load without breaking.✓ (2)

5.4 Limestone✓ and clay✓ (2)

5.5
- Cost saving
- Improved permeability and durability and durability of hardened concrete.
- Processed waste products from industrial processes
- Better bonding capability (Any 3 × 1) (3)

5.6
- Safety is guaranteed because the standards have been tested.
- Methods of construction described in the standards are cost effective.
- Standards are internationally accepted.
- Standards contain detailed guidelines and accurate calculations.
- Standards often provide alternative methods of producing the same results, leaving the user some choice. (Any 3 × 1) (3)

5.7 5.7.1
- Fertiliser contains chemicals and toxins that can stain and weaken the concrete.
- Fertiliser causes cracking and stalling once the concrete has set.

5.7.2
- Extra water will have to be added to the mix.
- Fine sand particles stick to the surfaces of larger ones, which prevent the cement from bonding to the aggregate.
- It weakens the concrete.
- It can produce cracks. (Any 2 × 2) (4)

5.8 The salts in calcium chloride reduce the strength of reinforcement✓ in the slab, which may cause failure of the structure.✓ (2)

5.9 100 mm–150 mm (1)
[25]

QUESTION 6

6.1
- Looks and behaves like mud
- Retains shape for a while when moulded into a ball
- Deformations not quick to close up or slump
- Deformations without any slump (Any 2 × 1) (2)

6.2
- Reducing the evaporation of moisture
- More workable and durable
- Increasing compressive strength
- Reducing water permeability
- Creating good bonding strength
- Prevents cracking of hardened mortar (Any 3 × 1) (3)

6.3 6.3.1 Moistening bricks✓ with water before they are laid✓

6.3.2 Mortar not required✓ for bond ing ✓ (2 × 2) (4)

6.4 Of a good grade and free from any organic matter (1)
[10]

QUESTION 7

7.1
- Plaster provides an additional barrier to water penetration.
- It is an effective method of increasing the rain resistance of a wall. (2)

7.2
- Hacking the surface
- Applying a spatter dash coat
- Bonding liquids
- Lathing
- Mechanically fitted mesh (Any 4 × 1) (4)

7.3
- Plaster mixes made with masonry cement will have lower strength than those with common cement at the same ratio of sand : cement.
- Plasters made with common cement are stronger and more impermeable than those made with masonry cement.
- Plasters made with made with masonry cement is less likely to crack because of shrinkage and slight movement of the substrate whereas the plaster mixes made with common cement are less flexible and more likely to crack.
- Masonry cement plasters resist sulphate attacks better than common cement plaster. (4)
[10]

QUESTION 8

8.1 Screeds are sand/cement mixes✓ that are not considered to contribute to the strength of the underlying slab.✓ (2)

8.2 1 : 4 (1)

8.3
- By placing a damp-proof membrane
- By placing insulation pad
- Applying a special paint (Any 2 × 1) (2)

8.4 Use two rigid blocks of identical height (15 mm).✓ A rigid, straight edge, 3 m long, is supported by the two blocks, and deviations from the level of the surface to the straight edge can be measured by using suitable feeler gauges✓ that can be slipped in and out of the space between the straight edge and the screed or topping layer.✓ (3)

8.5 Scatter a handful of coarse sand over an 800 mm² area and cover the particles with a transparent plastic sheet.✓ If condensation is visible after 24 hours, the floor is not yet ready for a moisture sensitive covering.✓ (2)
[10]

TOTAL: 100

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